

Multi-Reasoning Agent Decision Workflow

A 3-agent Langflow pipeline — Planner → Analyzer → Critic — that turns a raw decision scenario into a structured, self-critiqued Decision Memo.

Problem

A single LLM prompt asked to “recommend a decision” tends to skip steps: it jumps to a recommendation without decomposing the problem and never argues against its own output, making it easy to over-trust an untested memo.

Objective

Split the reasoning into three single-purpose agents so each step is auditable: **Planner** decomposes the decision (no recommendation); **Analyzer** commits to a recommendation with trade-offs and risks; **Critic** red-teams the memo and issues a verdict (APPROVE / REVISE / ESCALATE).

Approach & Trade-offs

- Sequential chain, not one mega-prompt — each agent only sees what its role needs, at the cost of 3 LLM calls instead of 1.
- No shared memory component — Langflow response → input_value edges pass the full upstream Message forward; each agent only has what was explicitly handed to it.
- Same model (GPT-4.1) across all three agents; max_iterations = 15 per agent as a hard safety cap.
- This is the 3-agent spine of a broader 12-step design (see docs/Alternate-Agent Instructions Library.pdf) — not the full pipeline.

Architecture

Chat Input → Planner Agent → Analyzer Agent → Critic Agent → Chat Output. All agents: OpenAI gpt-4.1, tool-calling enabled, API key resolved from the OPENAI_API_KEY env var (not hardcoded). See docs/diagram.svg for the full diagram.

Sample Data & Captured Results

Four NovaCart decision scenarios were drafted as test inputs (codebase/Prompts-Decision Briefs.pdf); three were run end-to-end in the Langflow Playground with the Critic agent's live verdict captured in docs/ProjectScreenshots.pdf:

Scenario	Verdict	Reason (from Critic agent output)
Premium Subscription Tier	REVISE	Needs finance sign-off and customer appetite probing before launch
Platform Migration	ESCALATE	No load testing / risk quantification — not ready for an informed decision
Vendor Selection	REVISE	Timeline too aggressive — needs cost projections, a decision rubric, fallback plan

Results / Learnings

- The Critic never defaulted to APPROVE across the three captured runs — findings were scenario-specific each time.
- The pipeline is strictly linear and stops at the Critic — REVISE/ESCALATE is a terminal output, with no automatic Reviser step.
- Keeping the Critic blind to the raw brief (it only sees the Analyzer's memo) is what makes its opposing-view argument genuinely independent.

Tech Stack

Langflow (flow exported at last_tested_version 1.10.2) · OpenAI GPT-4.1 via Langflow's Agent component · flow defined as exported Langflow JSON.

Full write-up: README.md · Diagram: docs/diagram.svg · Flow file: codebase/03-Langflow-MultiReasoningAgentWorkflow.json